

**GOVERNMENT GRADUATE COLLEGE OF
SCIENCE WAHDAT ROAD, LAHORE**



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Session

BS Chemistry (2019-23)

Assignment

Organic Chemistry [CHEM-317]

Topic

Past Paper 2020

Submitted To

Prof. Dr. Tariq Mahmood



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – 2020

Roll No.

Paper: Organic Chemistry
Course Code: CHEM-317 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following questions.

(5x4=20)

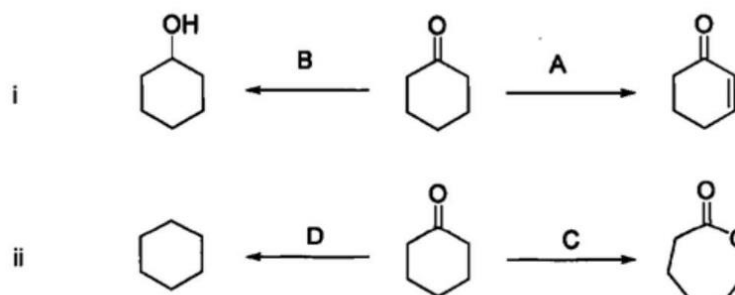
- I)** Explain Swern oxidation, give mechanism and example for oxidation of primary alcohols.
- II)** Write a short note on Birch reduction of benzaldehyde and methoxybenzene.
- III)** Describe two synthetic applications of free radicals.
- IV)** Describe types of vibration in IR spectroscopy.
- V)** Give four reactions with reagents to reduce aldehydes or ketones to hydrocarbons.

Q. No. 3.

- I)** Write a short note on ozonolysis of alkenes with mechanism and examples. **[5]**
- II)** Describe hydroboration of alkenes to yield alkanes and alcohols. **[5]**

Q. No. 4. Complete the following reactions and draw their mechanisms.

[2 × 5 = 10]



Q. No. 5. A) Write a note on applications of UV/Vis. spectroscopy. **[5]**

B) Give different methods for the generation of free radicals. **[5]**

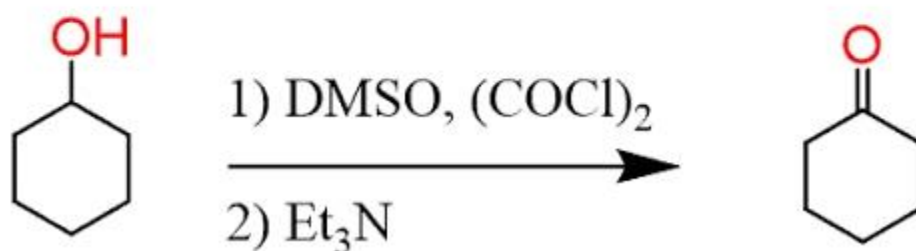
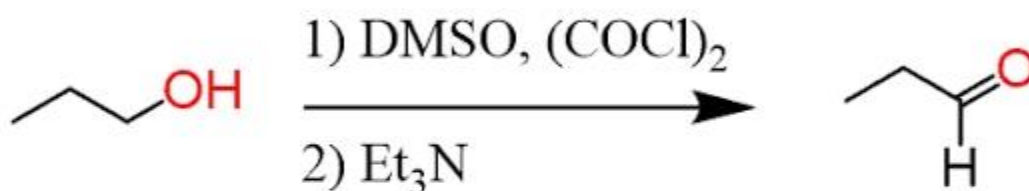
Q.1 . Explain swern oxidation , give mechanism and an example for oxidation of alcohals.

Ans:

The Swern oxidation named after Daniel Swern, is a chemical reaction whereby a primary or secondary alcohol is oxidized to an aldehyde or ketone using oxalyl chloride, dimethyl sulfoxide (DMSO) and an organic base,

Such as triethylamine. It is one of the many oxidation reactions commonly referred to as 'activated DMSO' oxidations. The reaction is known for its mild character and wide tolerance of functional groups.

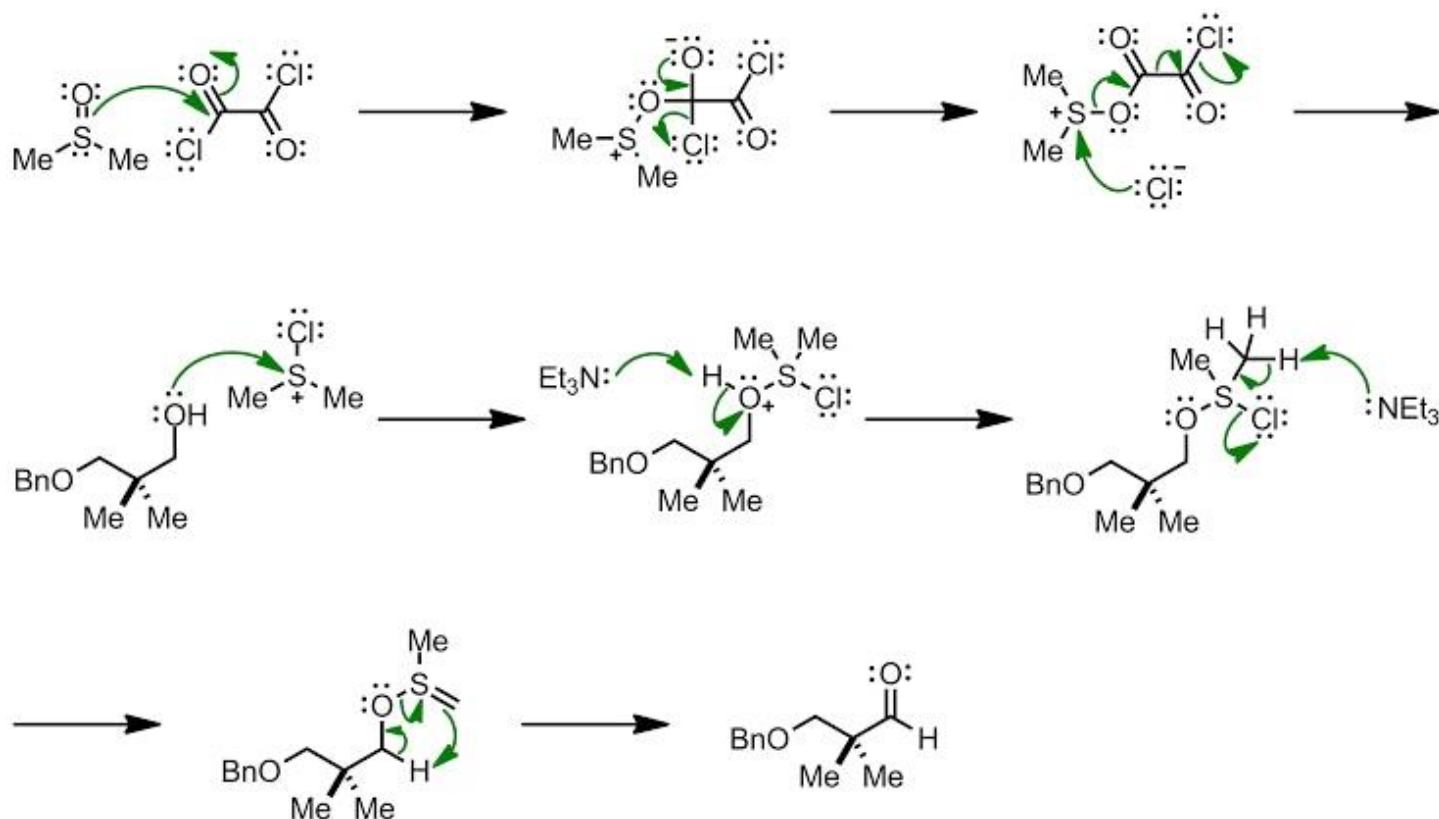
Swern Oxidation: 1° Alcohols - Aldehyde, 2° - Ketone



Mechanism:

The first step of the Swern oxidation is the low-temperature reaction of DMSO, formally as resonance contributor with oxalyl chloride. 2. The first intermediate quickly decomposes giving off carbon dioxide and carbon monoxide and producing chloro(dimethyl)sulfonium chloride.

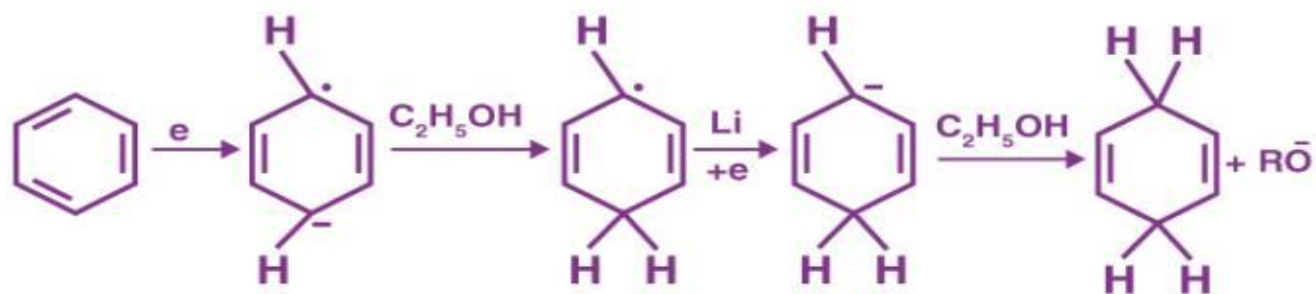
After addition of the alcohol, the chloro(dimethyl)sulfonium chloride reacts with the alcohol to give the key alkoxy-sulfonium ion intermediate. The addition of at least 2 equivalents of base — typically triethylamine — will deprotonate the alkoxy-sulfonium ion to give the sulfur ylide. In a five-membered ring transition state, the sulfur ylide decomposes to give dimethyl sulfide and the desired carbonyl compound.



Q. 2 .Write a note on Birch reduction of Benzaldehyde and methoxybenzene.

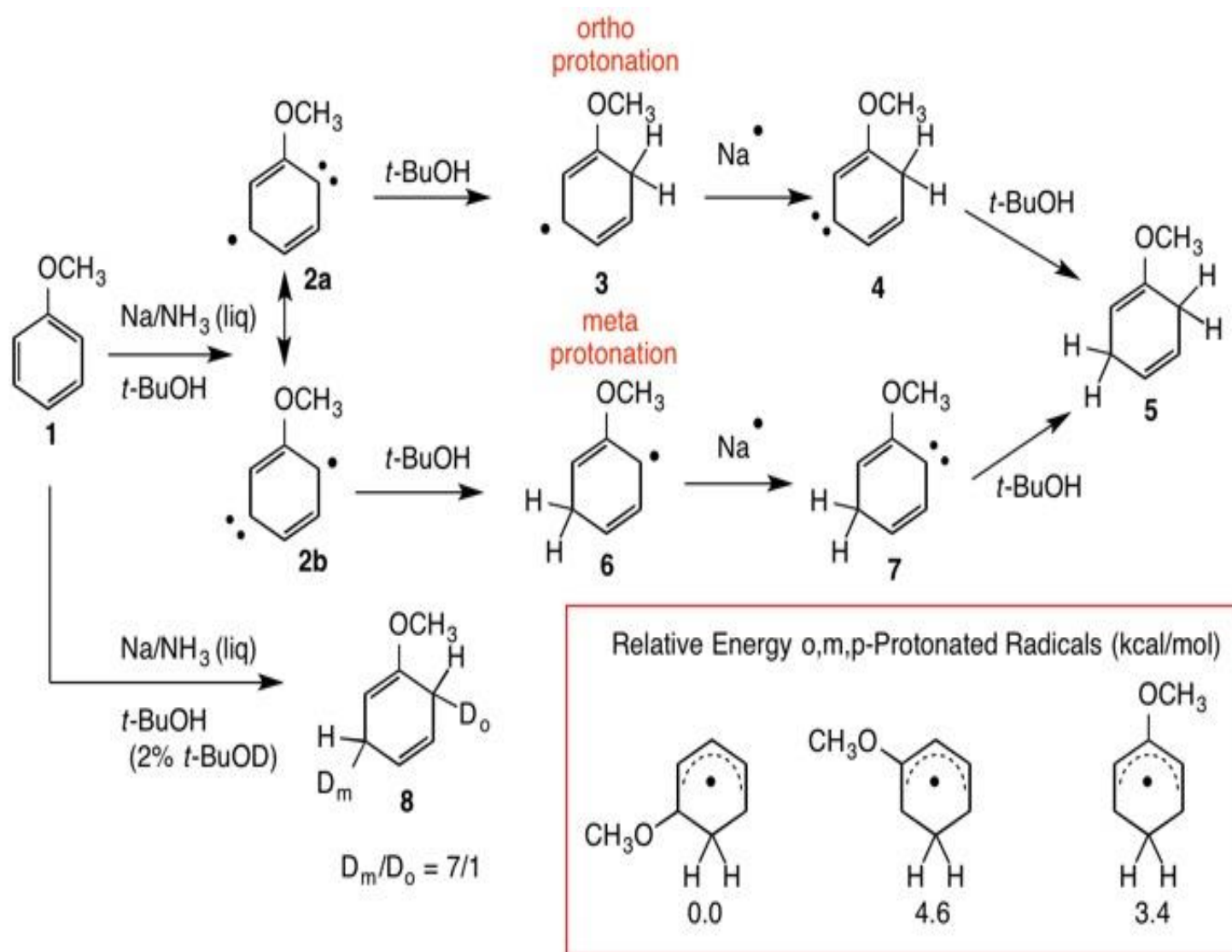
Ans:

The Birch reduction is an organic chemical reaction where aromatic compounds which have a benzenoid ring are converted into 1,4-cyclohexadiene which have two hydrogen atoms attached at opposite ends of the molecule. It is a very useful reaction in synthetic organic chemistry.

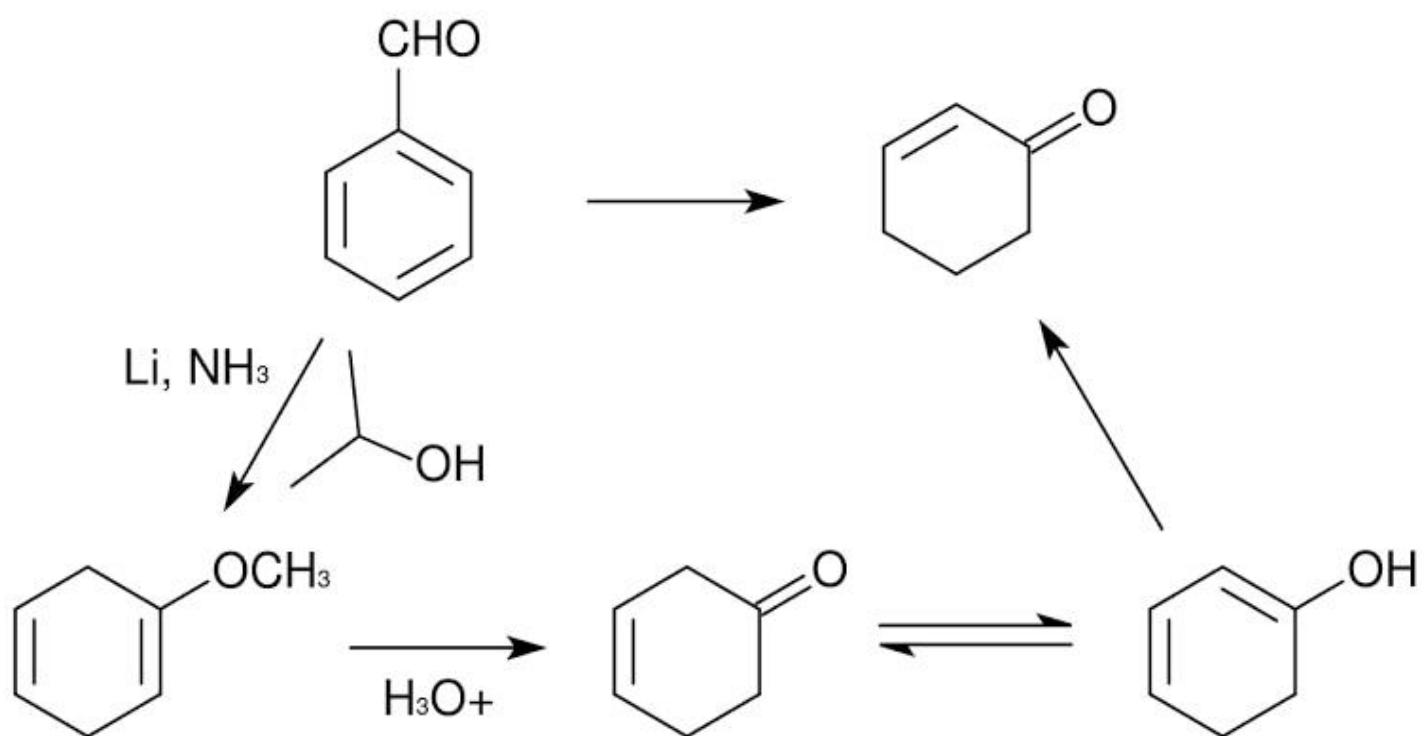


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Reduction of Methoxybenzene.



Reduction of Benzaldehyde:

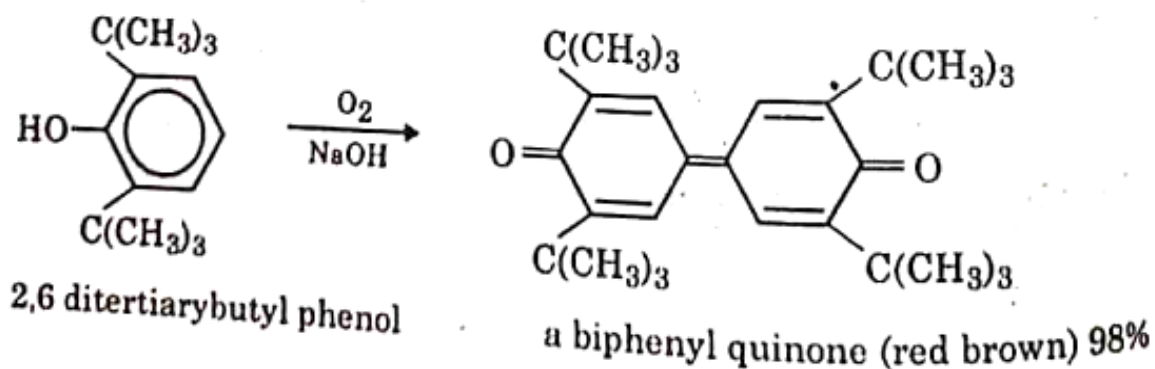


Q. 3. Describe two synthetic applications of free radicals?

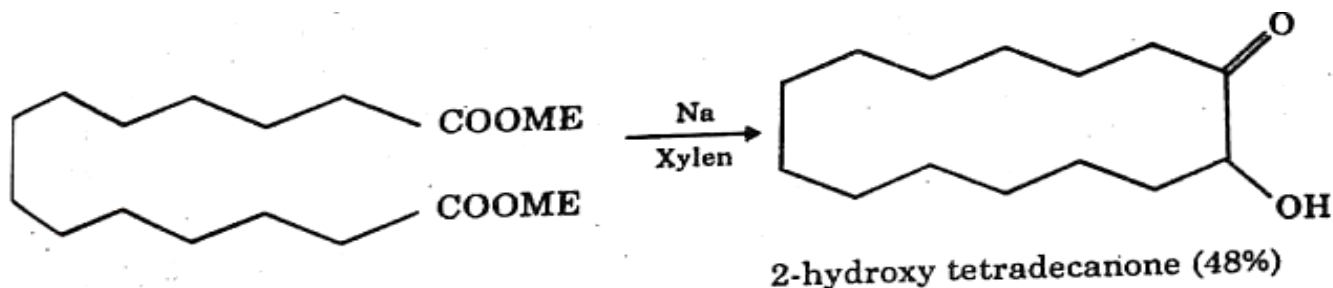
Ans:

Applications free radicals are

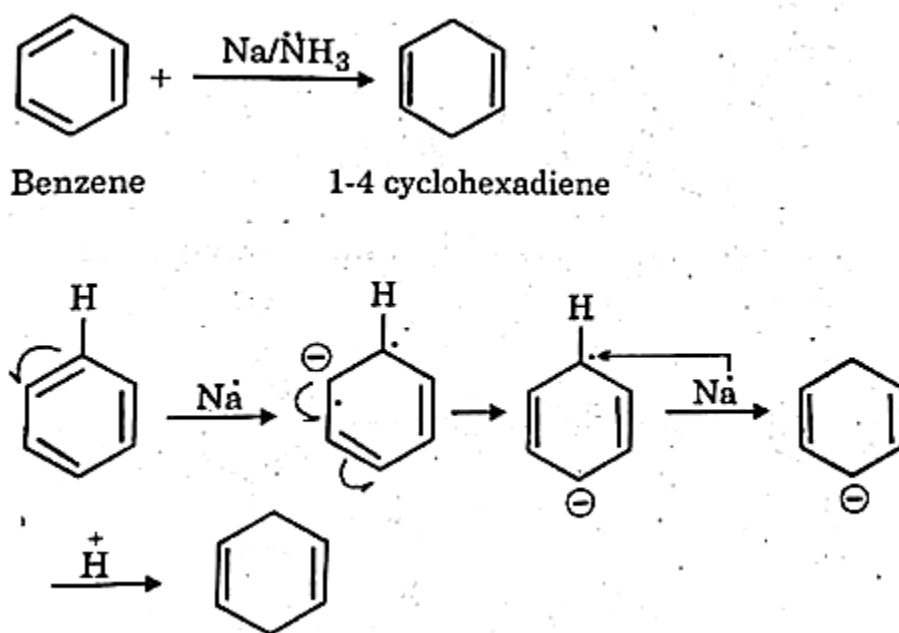
- (i) Oxidative coupling of phenols are used for **synthesis** of Drugs and Dyes.



- (ii) The acyloin condensation is one of the best method to use when synthesis medium to large size ring.



- (iii) Redical reduction reaction generally involves the addition of hydrogen to pi-bond. Sodium or Lithium methyl in liquid NH_3 or alcohol is used a reducing agent.



Q. 4. Describe types of vibration in IR spectroscopy.

Ans:

There are two majors types of vibration in IR spectroscopy that has further modes of vibrations .

These types are:

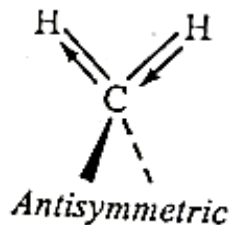
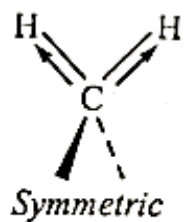
- ✚ Stretching Vibrations
- ✚ Bending vibrations

Stretching Vibrations:

When there is a continuous change in the interatomic distance along the axis of the bond between two atoms, this process is known as a stretching vibration.

In AX_2 system, This type has two modes of vibrations that are :

- Symmetric
- Anti symmetric

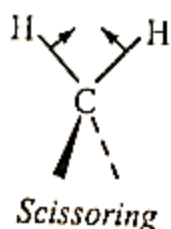


Stretching vibrations

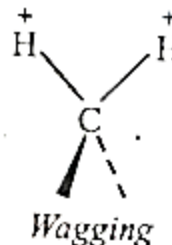
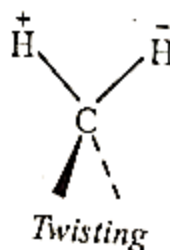
Bending Vibrations:

A change in the angle occurring between two bonds is known as a bending vibration. Four bending vibrations exist namely.

- Wagging
- Twisting
- Rocking
- Scissoring



In-plane



Out-of-plane

Bending vibrations

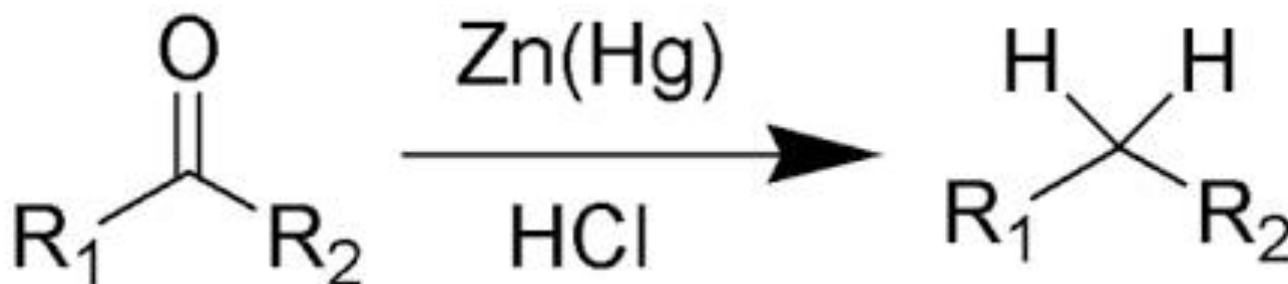
Q. 5. Give four reaction with reagents to reduce aldehyde or ketones to hydrocarbons.

Ans:

The conversion of aldehydes or ketones to hydrocarbons can be carried out by various reagents. Out of which four are given follow:

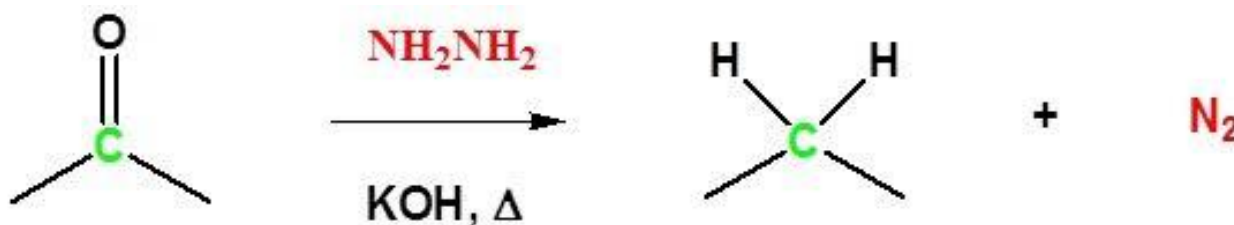
Clemmensen Reduction

The carbonyl group of aldehydes or ketones changes to methyl or methylene when reacts with zinc amalgam in presence of HCl.



Wolff-Kishner Reduction

The Wolff-Kishner reduction is the reduction of carbonyl compounds to hydrocarbons by heating them with hydrazine hydrate and a base.

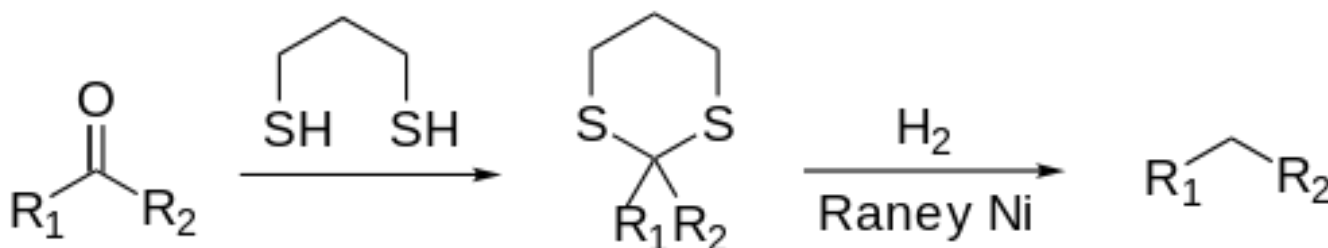


Aldehyde or
Ketone

Alkane

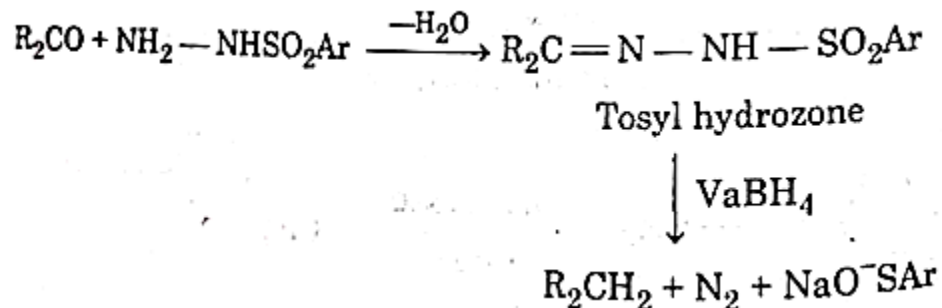
Mozingo reaction

The reduction works for both aldehydes and ketones and involves two steps. First, the carbonyl compound is converted into a thioketal (or thioacetal). In a second step, the thioketal is subsequently reduced to the corresponding methylene compound.



Tosylhydrozone Method:

Carbonyl compound is converted into Tosylhydrozone with toluene-*p* sulphonylhydrazine which is reduced with reduced with metal hydride.



LONG QUESTIONS

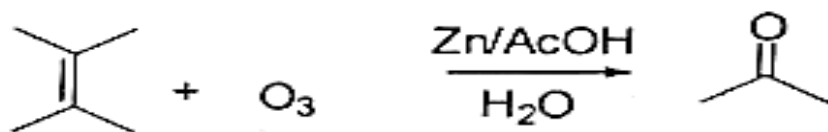
Q.3 (I). Write a short note on ozonolysis of alkenes, with mechanism and example.

Ans:

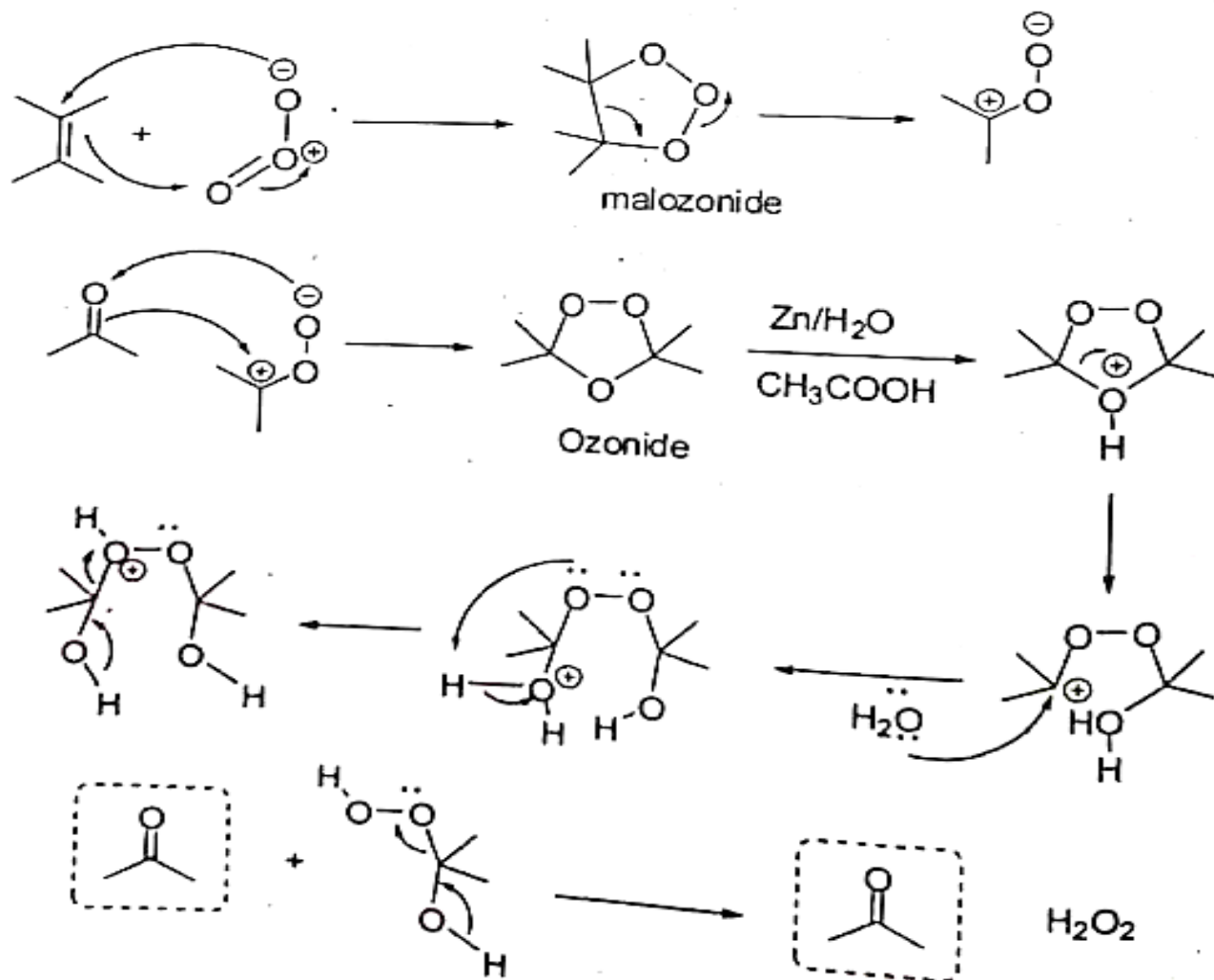
Ozone reacts with olefins forming primary ozonide which is also called Malozonide . Malozonide undergo rearrangement to form isolable ozonide.

The ozonide work up with Zn/AcOH to give two carbonyl compounds.

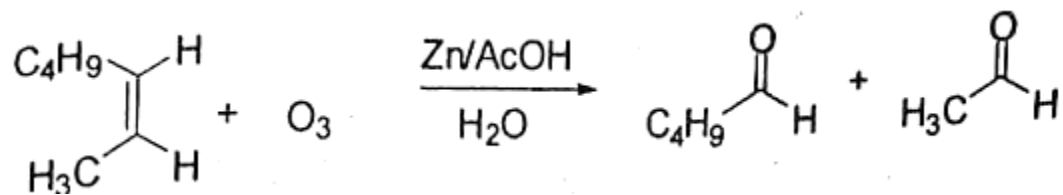
Reaction:



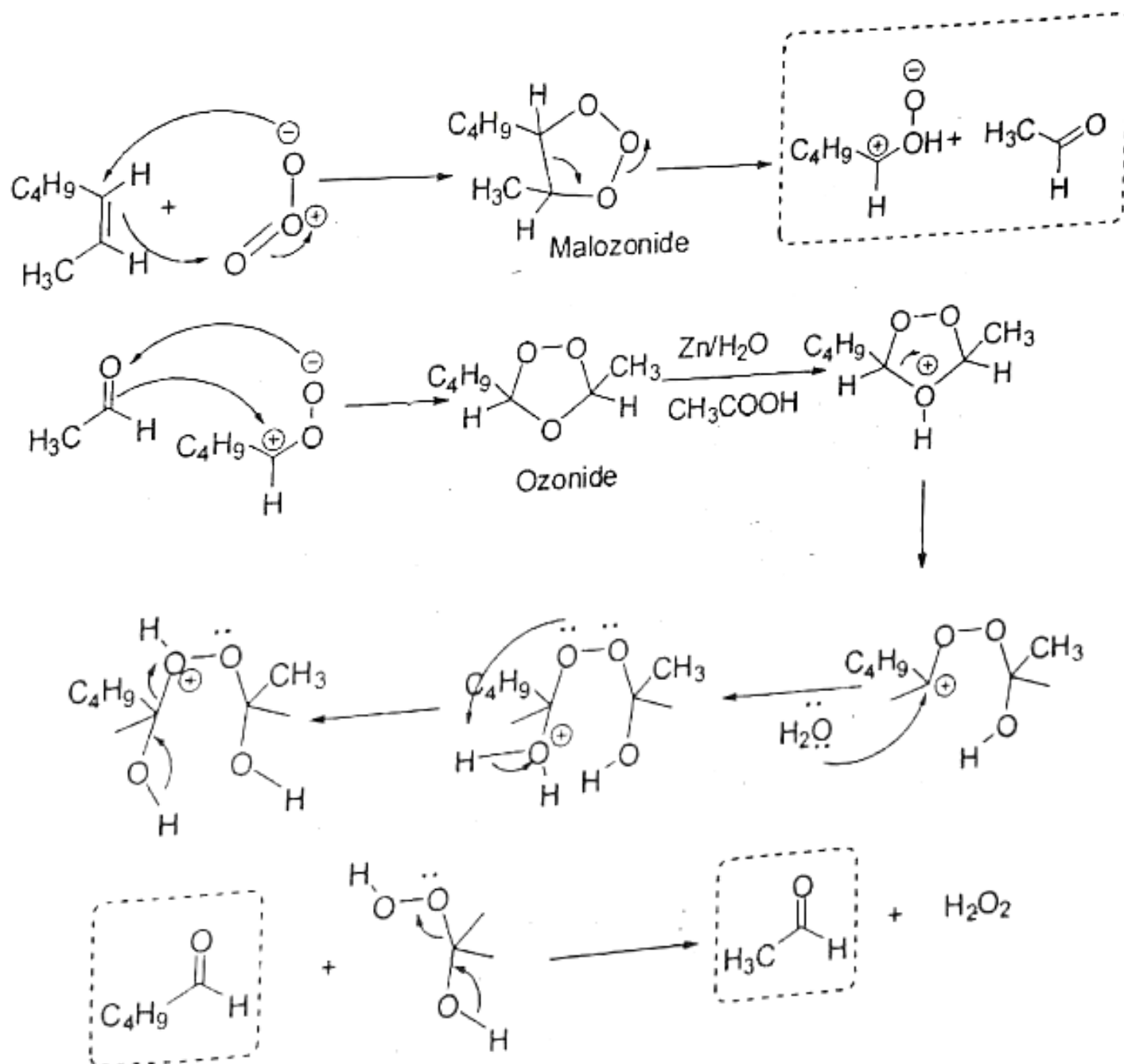
Mechanism:



EXAMPLE



MECHANISM



Q.3. (II). Describe hydroboration of alkene to yield alkane and alcohols.

Ans:

Hydroboration reaction:

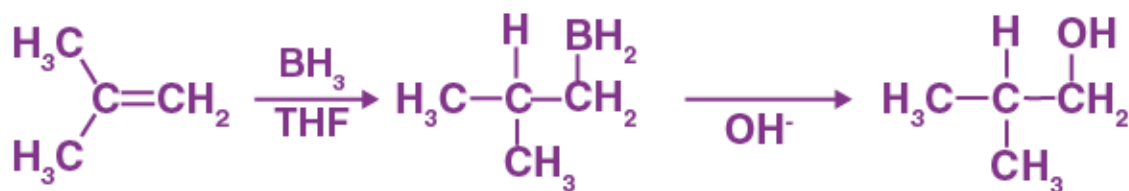
The Hydroboration reaction is an organic chemical reaction which is employed for the conversion of alkenes into alcohols or alkanes that are neutral.

Hydroboration to Alcohol.

The Hydroboration Oxidation reaction is an organic chemical reaction which is employed for the conversion of alkenes into alcohols that are neutral. This is done via a two-step process which includes a hydroboration step and an oxidation step. This is done by a net addition (across the entire double bond) of water.

The hydroboration oxidation reaction mechanism can be considered as an anti-Markovnikov reaction where a hydroxyl group attaches itself to the carbon which is less substituted. The first time the Hydroboration oxidation reaction was reported was in the second half of the 1950s by the English born American chemist Herbert Charles Brown. He went on to win the Nobel Prize in Chemistry in the year 1979 for this work.

Reaction:



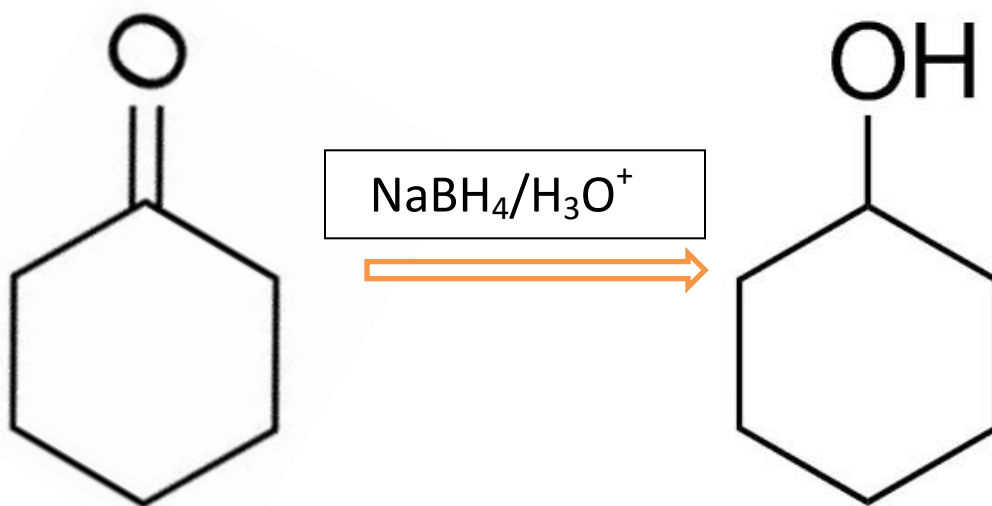
Alkane to alkene:

The reaction is also known as the hydrogenation of alkene. The reaction occurs in the presence of a finely divided metal catalyst such as nickel (Ni), palladium (Pd), platinum (Pt), or rhodium (Rh) at a temperature of about 150°C

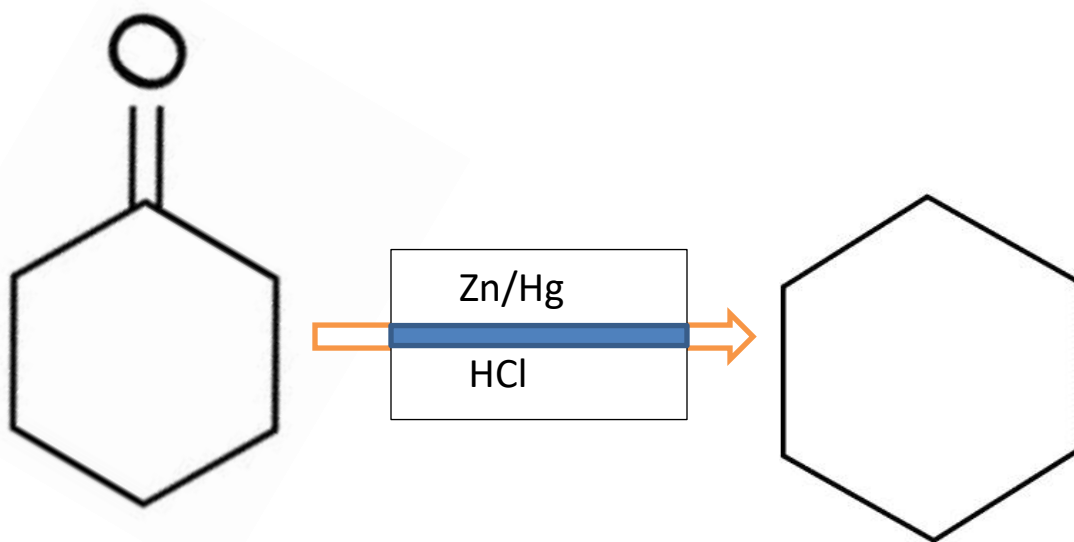
Q. 4 . Complete the Following reaction?

Ans:

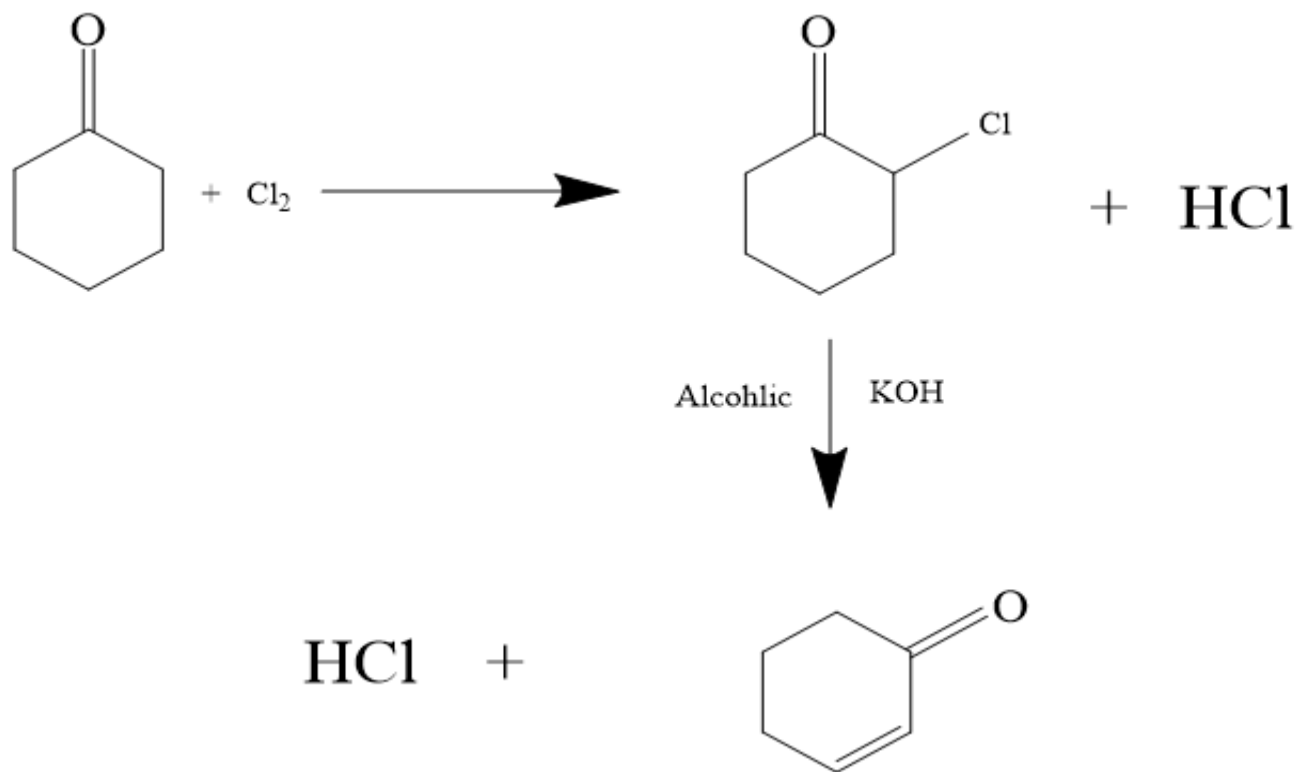
Part B



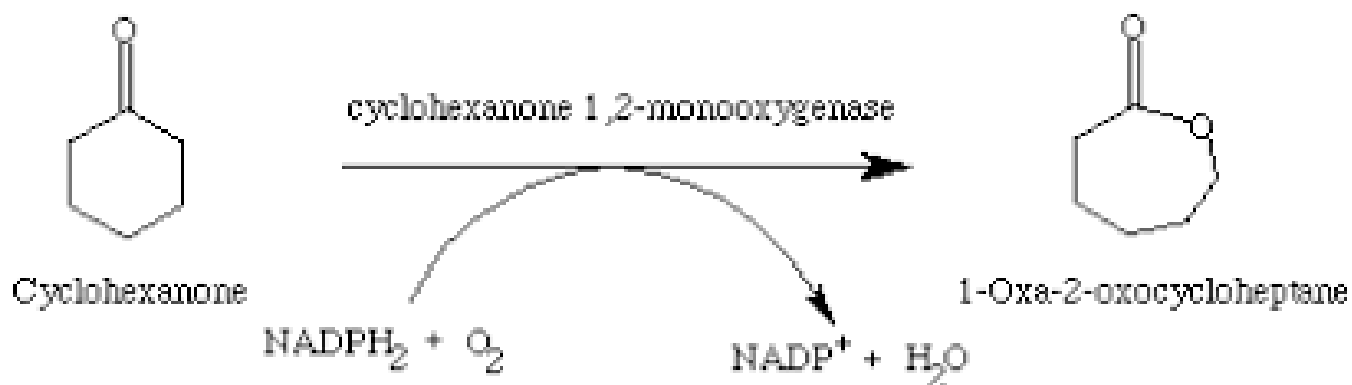
Part D



Part A



Part C



Q. 5. Applications of UV/ Visible Spectroscopy?

Ans:

UV/ Visible Spectroscopy has two Major types of applications:

- ❖ Quantitative
- ❖ Qualitative

Although various applications of UV/ Visible Spectroscopy are:

1. Detection of Impurities

UV absorption spectroscopy is one of the best methods for determination of impurities in organic molecules. Additional peaks can be observed due to impurities in the sample and it can be compared with that of standard raw material. By also measuring the absorbance at specific wavelength, the impurities can be detected. Benzene appears as a common impurity in cyclohexane. Its presence can be easily detected by its absorption at 255 nm.

2. Structure elucidation of organic compounds

UV spectroscopy is useful in the structure elucidation of organic molecules, the presence or absence of unsaturation, the presence of hetero atoms. From the location of peaks and combination of peaks, it can be concluded that whether the compound is saturated or unsaturated, hetero atoms are present or not etc.

3. Chemical kinetics

Kinetics of reaction can also be studied using UV spectroscopy. The UV radiation is passed through the reaction cell and the absorbance changes can be observed.

4. Qualitative analysis

UVabsorption spectroscopy can characterize those types of compounds which absorbs UV radiation. Identification is done by comparing the absorption spectrum with the spectra of known compounds. UV absorption spectroscopy is generally used for characterizing aromatic compounds and aromatic olefins.

5. Quantitative analysis

UV absorption spectroscopy can be used for the quantitative determination of compounds that absorb UV radiation. This determination is based on Beer's law.

Q. 5(II) . Describe different methods for generation of free radicals.

Ans:

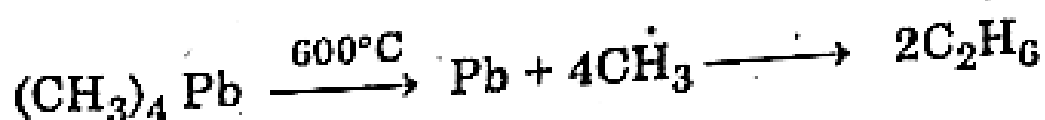
There are two types free radicals on the basis of there life;

- ❖ Short lived free radicals
- ❖ Long lived/ Stable free radicals

The methods used for generation of free radicals are.

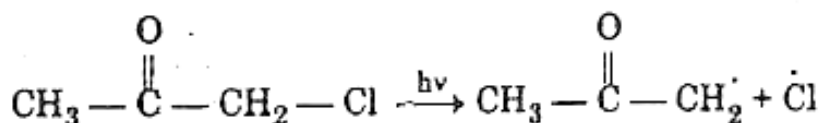
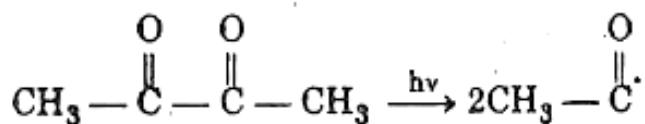
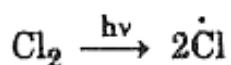
Thermolysis of weak bonds.

Compound containing weak bonds gives free radicals on heating like lead or mercury alkyls.



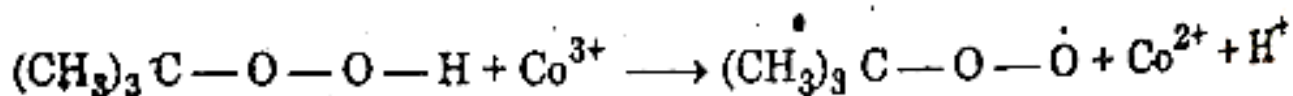
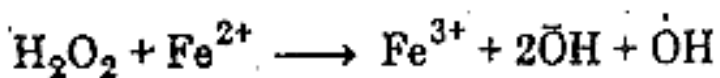
Photolysis.

Absorption of UV and visible light provides sufficient energy to break weak bond homolytically



Redox Reaction.

Redox reaction with inorganic ions that can change their valence state by gain or loss of single electron produce free radicals.



<Thank you>